



USTP FINAL PROJECT

PROJECT: “DESIGN AND UVM BASED VERIFICATION OF AXI TO APB BUS PROTOCOL”

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PHASE 01: PROJECT FOUNDATION (REPORT)

Deliverables:

- Top Level Block Diagram
- Subblock Specifications

1. INTRODUCTION:

The Design and UVM Based Verification of AXI to APB Bus Protocol project focuses on designing a bridge that enables communication between an AXI-based system and APB-based peripherals. The bridge is responsible for translating and managing transactions between the two bus protocols while maintaining proper control, data transfer, and response handling. The project also includes developing a UVM-based verification environment to verify the functionality and reliability of the designed bridge through systematic testing and coverage.

2. PROJECT OBJECTIVES:

The main objectives of this project are:

- To design an AXI-to-APB bridge for communication between AXI-based systems and APB-based peripherals.
- To understand and implement the basic transaction flow between the AXI and APB protocols.
- To develop RTL logic for managing data transfer, control, and responses between the two interfaces.
- To verify the bridge functionality using a structured UVM-based verification environment.
- To evaluate the design through simulation and functional verification to improve its reliability and correctness.

3. SYSTEM OVERVIEW:

The proposed system consists of an AXI-to-APB bridge that enables communication between an AXI-based system and APB-based peripherals. The bridge receives transactions from the AXI side, processes and translates them into the required APB operations, and manages the corresponding responses back to the AXI side. The overall system can be represented as:

AXI Master → AXI Interface → AXI-to-APB Bridge → APB Interface → APB Slave

The bridge forms the central component of the system and is responsible for coordinating address, data, control, and response information between the two protocols. The design will be further developed and verified through RTL simulation and a UVM-based verification environment.

4. TOP-LEVEL BLOCK DIAGRAM:

4.1. BLOCK DIAGRAM:

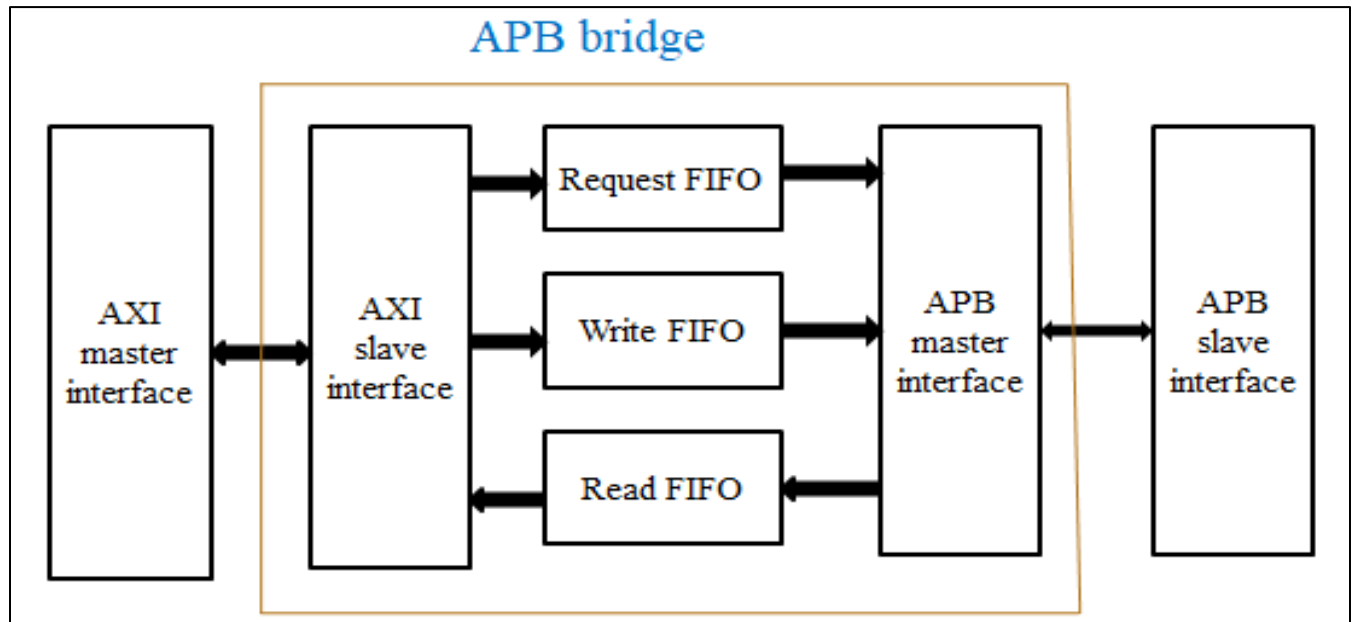


Figure 1: BASIC ARCHITECTURE OF AXI TO APB BRIDGE

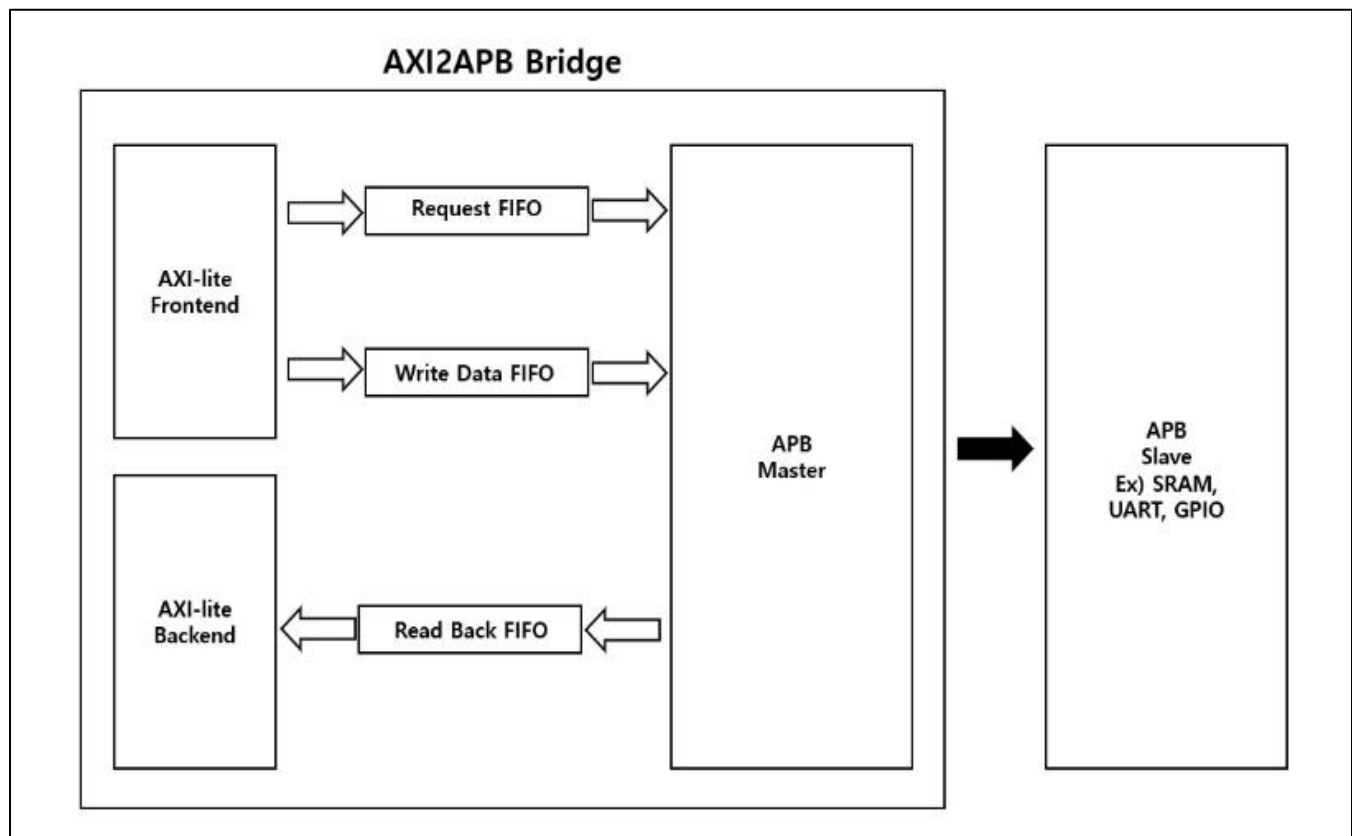


Figure 2: OVERALL BLOCK DIAGRAM AXI TO APB

4.2. BLOCK DIAGRAM DESCRIPTION:

The proposed system is organized around an AXI-to-APB bridge, which acts as an interface between an AXI-based master and an APB-based peripheral. The major blocks and their roles are described below:

- **AXI Master:** Initiates read and write transactions and communicates with the bridge through the AXI interface.
- **AXI Interface:** Receives and manages AXI transaction information, including address, data, control, and response signals.
- **AXI-to-APB Bridge:** Processes the incoming AXI transaction and translates it into the corresponding APB operation.
- **APB Interface:** Generates the required APB signals and manages communication with the connected APB peripheral.
- **APB Slave:** Receives the APB transaction, performs the requested operation, and provides the required response or data back through the bridge.

Overall, the bridge coordinates the transaction flow between the two protocols, providing a clear path for address, data, control, and response information from the AXI side to the APB side and back.

5. SUBBLOCK SPECIFICATIONS:

AXI to APB Bridge – Subblocks Specification

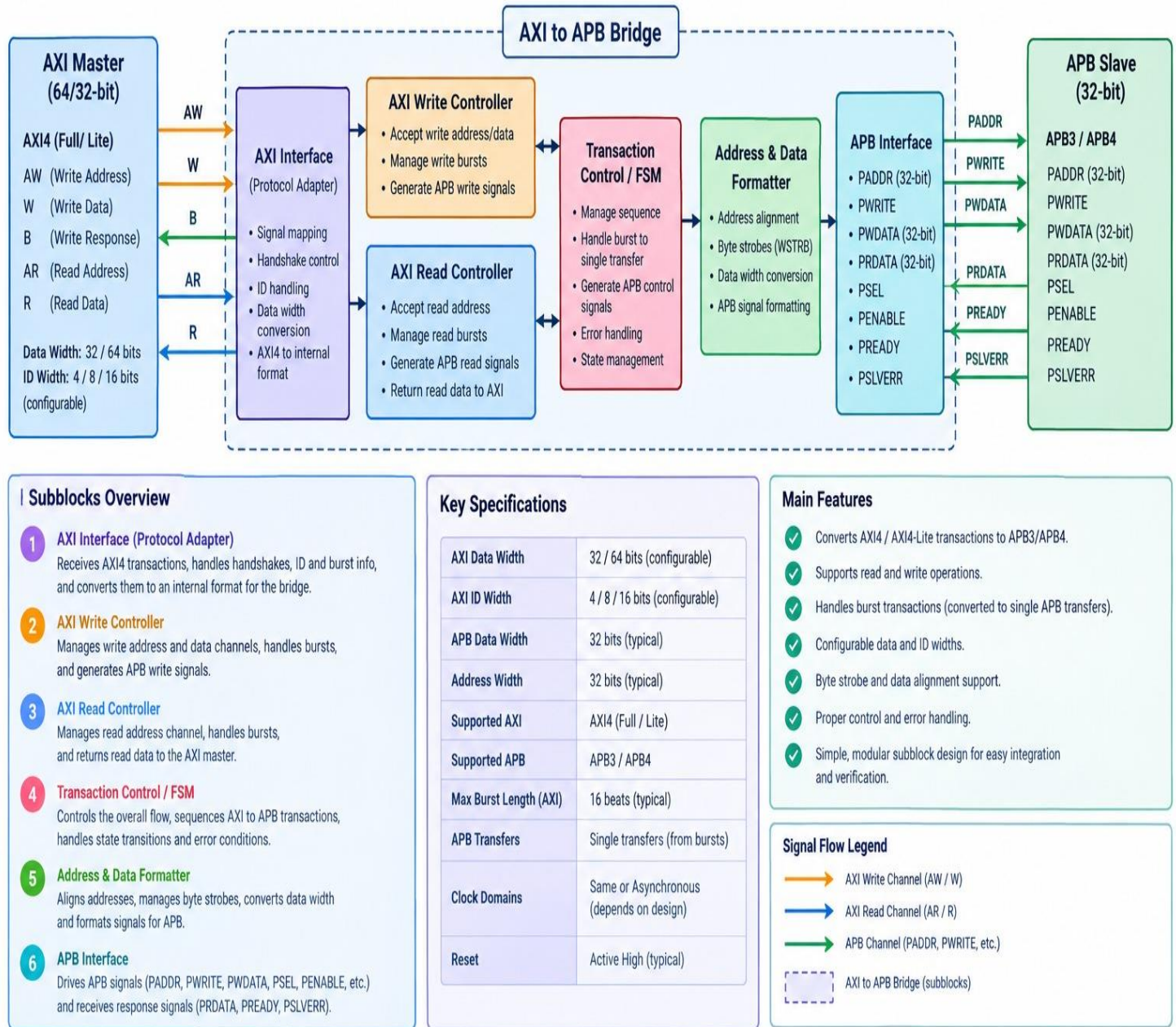


Figure 3: AXI TO APB SUBBLOCKS SPECIFICATION

6. INTERFACE SIGNAL SPECIFICATION:

The AXI-to-APB bridge communicates with the AXI and APB interfaces through dedicated signals for **clock, reset, address, data, control, and response handling**. The main signals are summarized below.

6.1. AXI INTERFACE SIGNALS:

The AXI side provides separate channels for read and write transactions. Each channel uses a VALID/READY handshake to control the transfer of information.

Signal Group	Signals	Purpose
Clock & Reset	ACLK, ARESETn	Provide clocking and reset control
Write Address	AWADDR, AWVALID, AWREADY	Transfer the write address
Write Data	WDATA,WSTRB, WVALID, WREADY	Transfer write data and byte-enable information
Write Response	BRESP, BVALID, BREADY	Provide the write transaction response
Read Address	ARADDR, ARVALID, ARREADY	Transfer the read address
Read Data	RDATA, RRESP, RVALID, RREADY	Return read data and response information

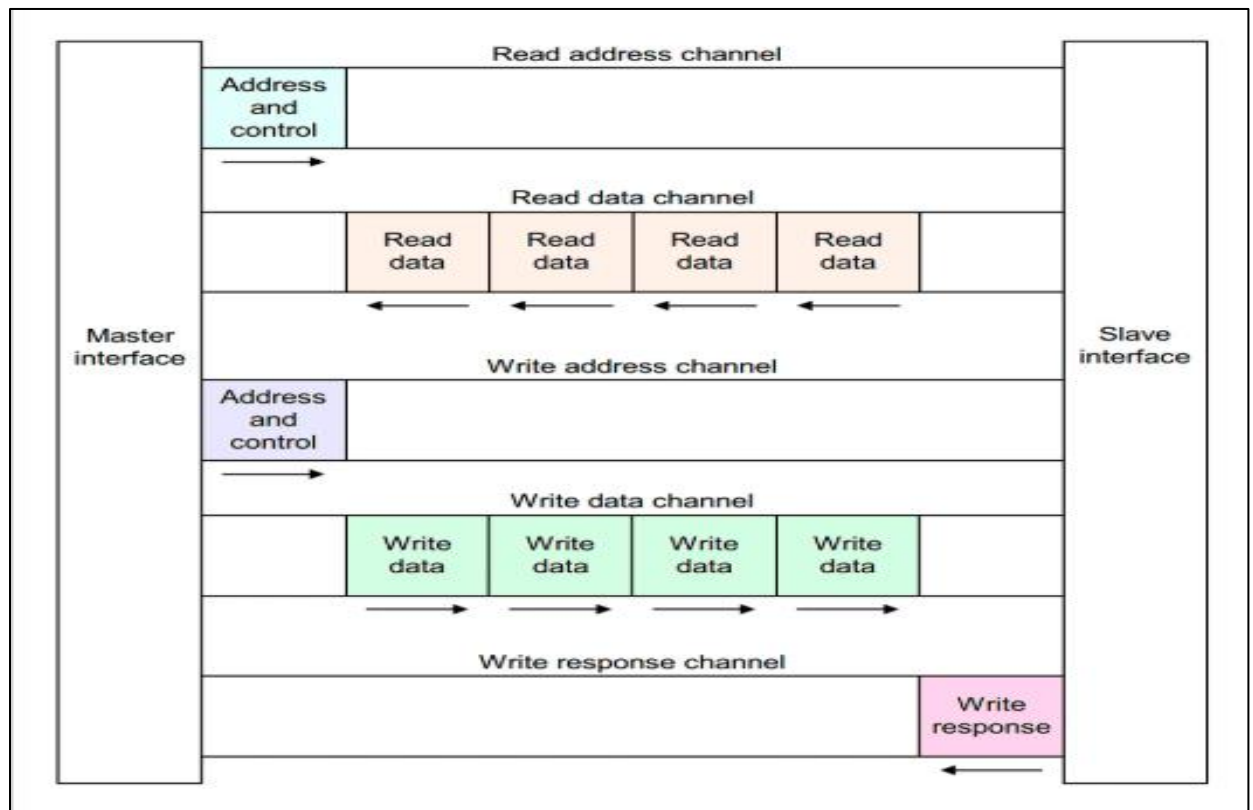


Figure 4: AXI MASTER AND SLAVE CONNECTED VIA THE FIVE AXI CHANNELS

AXI TO APB BUS PROTOCOL

6.2. APB INTERFACE SIGNALS:

The APB side uses a simple two-phase transaction mechanism consisting of SETUP and ACCESS phases.

Signal Group	Signals	Purpose
Clock & Reset	PCLK, PRESETn	Provide clocking and reset control
Address	PADDR	Specifies the APB address
Control	PWRITE, PSEL, PENABLE	Control the APB transaction and its phases
Write Data	PWDATA	Carries data for write operations
Read Data	PRDATA	Carries data from the APB slave during reads
Status	PREADY, PSLVERR	Indicates transaction completion and error status

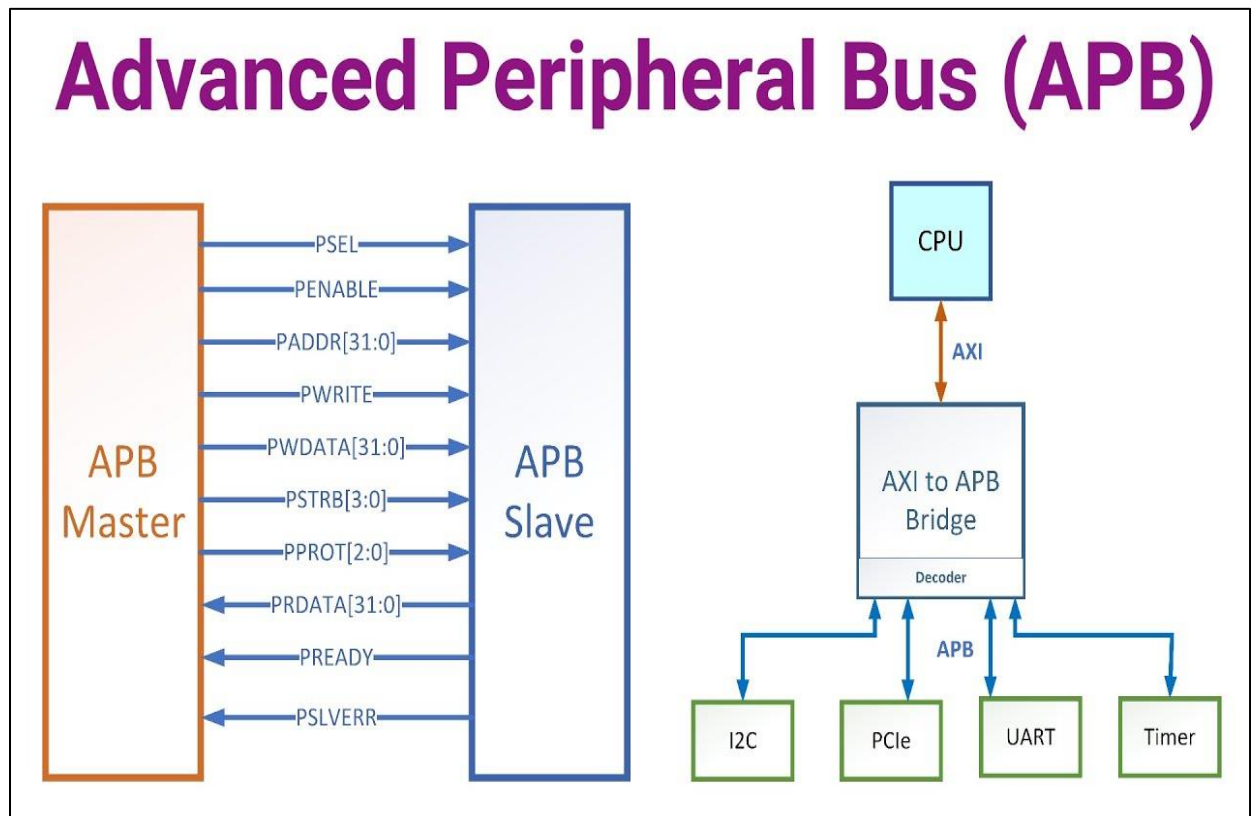


Figure 5: ADVANCED PERIPHERAL BUS INTERFACE SIGNALS

7. BASIC TRANSACTION FLOW:

The AXI-to-APB bridge manages the transfer of read and write transactions between the AXI master and APB slave. The general transaction flow is described below.

7.1. AXI WRITE TRANSACTION:

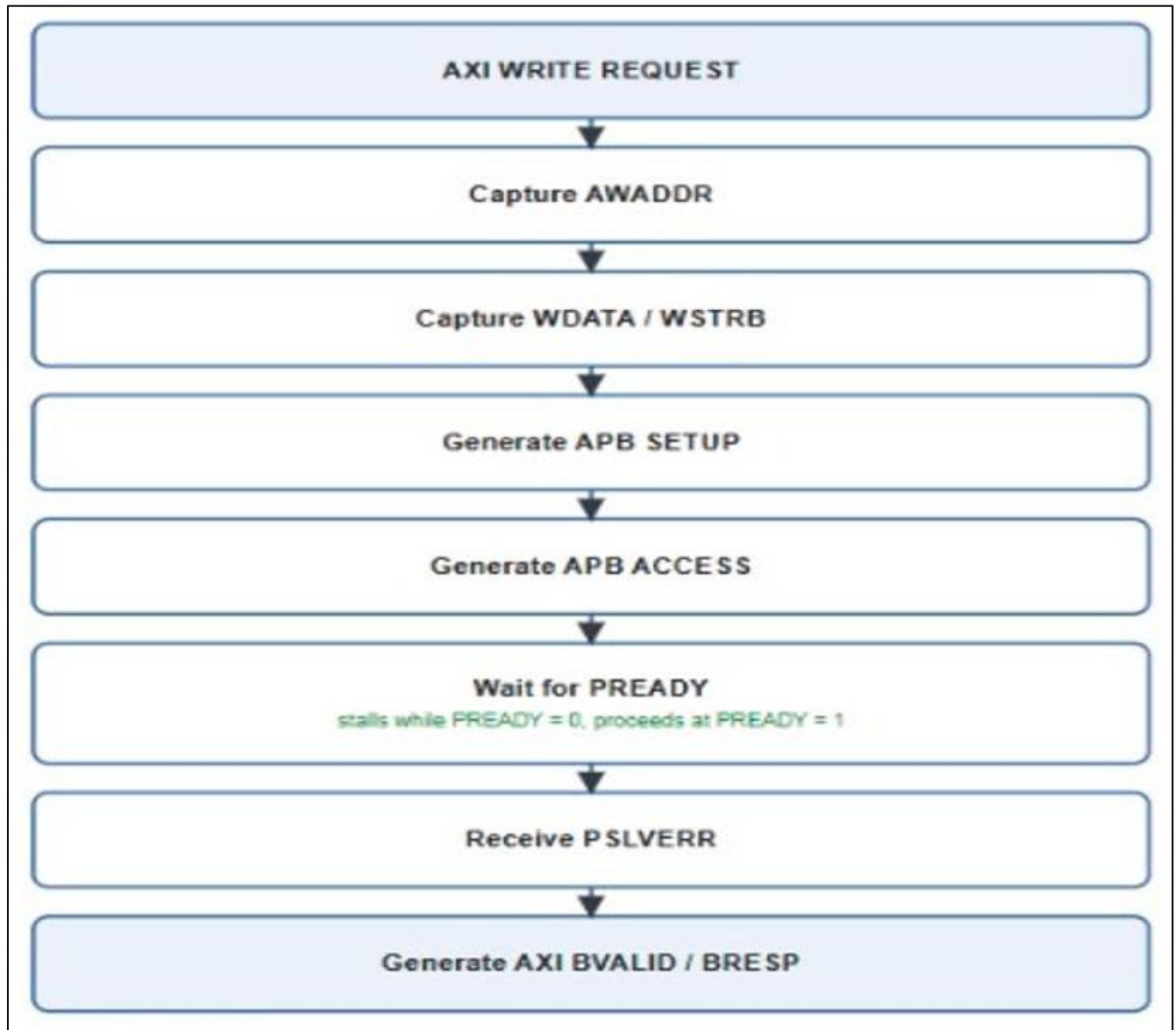


Figure 6: AXI WRITE TRANSACTION FLOW

7.2. AXI READ TRANSACTION:

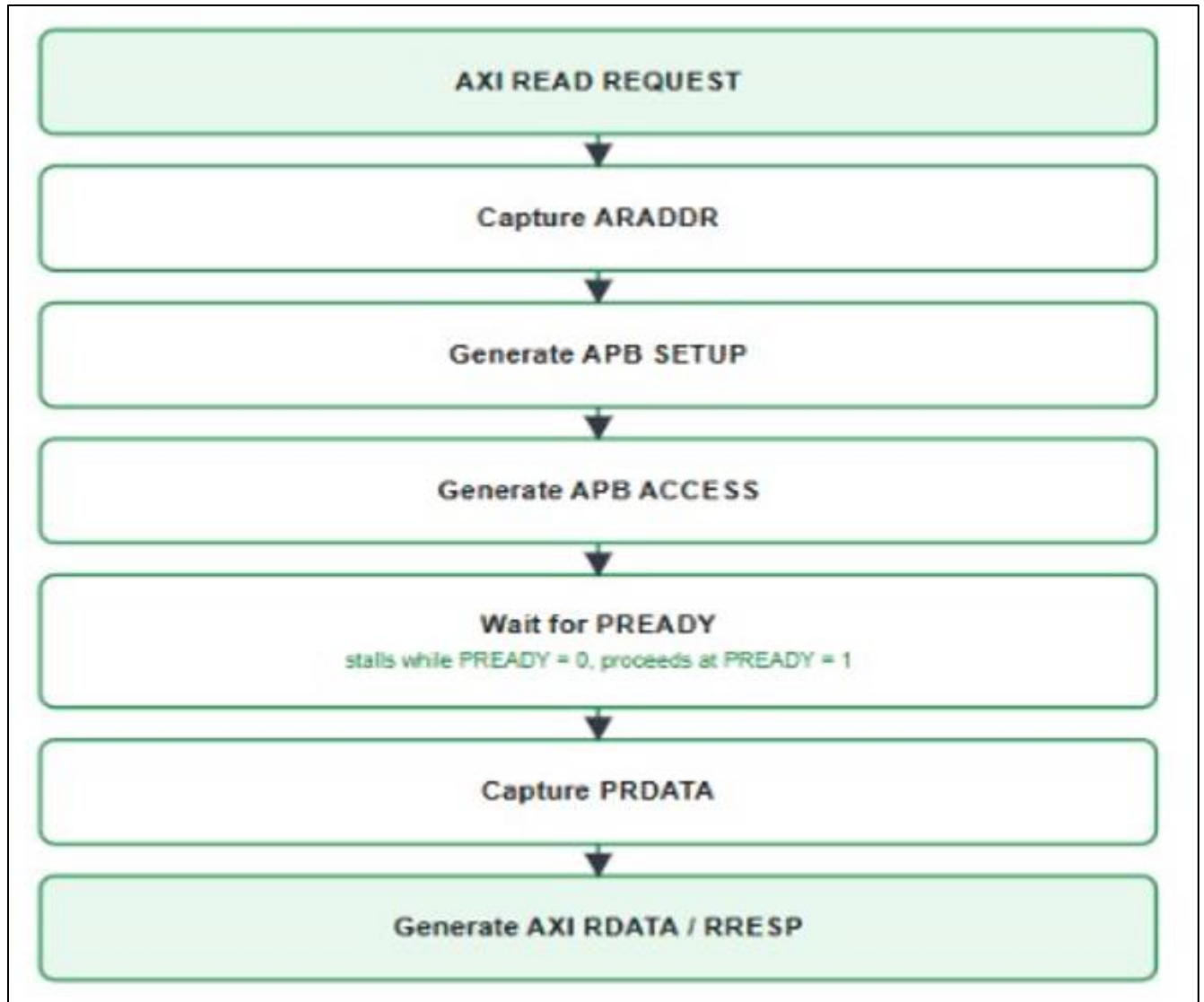


Figure 7: AXI READ TRANSACTION FLOW

8. CONCLUSION:

This phase report established the basic foundation of the AXI-to-APB Bus Protocol project by defining its overall architecture, major functional blocks, interfaces, and basic transaction flow. The AXI-to-APB bridge provides a structured way to manage communication between AXI-based systems and APB-based peripherals. These specifications will serve as a foundation for the upcoming RTL design, simulation, and UVM-based verification of the project.